

Supplementary Material for RG-EditFlow: Reliability-Gated Edit Flow for Offline Model-Based Optimization

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Appendix A: Design-Bench Task Details

Design-Bench supplies a fixed offline dataset for each task and reserves the task oracle or evaluator for post-hoc assessment of generated candidates. In our experiments, all training and selection signals are computed without querying the oracle on new designs; the oracle is used only after the final candidate set has been fixed.

Table 1: Design-Bench task details used in the experiments. Offline size is the number of design–score pairs available for training; category size applies only to discrete sequence tasks.

Task	Offline size	Dim.	Cat.	Type	Oracle/evaluator and objective
Ant	10,004	60	–	Continuous	Exact simulator; maximize locomotion performance of the ant morphology.
D’Kitty	10,004	56	–	Continuous	Exact simulator; maximize movement/control performance of the D’Kitty morphology.
Superconductor	17,014	86	–	Continuous	Random-forest evaluator; maximize predicted superconducting critical temperature.
TFBind8	32,898	8	4	Discrete	Exact lookup oracle; maximize transcription-factor DNA binding affinity.
TFBind10	10,000	10	4	Discrete	Exact lookup oracle; maximize transcription-factor DNA binding affinity.

Continuous morphology and materials tasks. Ant and D’Kitty evaluate continuous morphology vectors in simulator-based robot-control environments. These tasks test whether local edits from elite seeds can improve physical designs without drifting into unsupported regions of morphology space. Superconductor uses continuous materials features and a random-forest evaluator for critical temperature, so nearest-neighbor and displacement filters are computed in the benchmark feature representation rather than in a simulator morphology representation.

Discrete sequence tasks. TFBind8 and TFBind10 are DNA sequence-design tasks over four nucleotide categories. We treat them as embedded discrete design spaces during proposal construction, then evaluate only valid decoded sequences under the benchmark lookup oracle. This distinction

matters for the method: the flow proposes local edits in the representation used for optimization, while final benchmark scores are assigned to the decoded candidate sequences.

Appendix B: Candidate Distribution Summary

All reported scores are normalized by Design-Bench oracle bounds, and all aggregate entries use eight seeds unless stated otherwise. The main paper focuses on normalized 100th-percentile performance for compact comparison. To show whether a method improves the broader generated candidate set rather than only finding a single strong endpoint, Tables 2 and 3 report the 80th- and 50th-percentile Design-Bench results under the same evaluation protocol. The 80th percentile reflects the quality of the upper portion of the generated candidate set, while the median measures whether useful candidates are produced consistently.

Table 2: Comparison with representative prior baselines on Design-Bench in terms of the normalized 80th-percentile score. The best and second-best results are shown in bold and underlined, respectively.

Method	Ant	D’Kitty	Superconductor	TFBind8	TFBind10
D_{best}	0.565	0.884	0.400	0.439	0.467
BO-qEI	0.629 ± 0.000	0.884 ± 0.000	0.325 ± 0.001	0.439 ± 0.000	0.510 ± 0.011
CMA-ES	0.007 ± 0.013	0.718 ± 0.001	0.389 ± 0.004	0.652 ± 0.017	0.543 ± 0.013
REINFORCE	0.182 ± 0.017	0.562 ± 0.197	0.364 ± 0.005	0.622 ± 0.030	0.519 ± 0.007
COMs	0.635 ± 0.031	0.887 ± 0.004	0.328 ± 0.124	0.738 ± 0.027	0.526 ± 0.012
CbAS	0.542 ± 0.034	0.813 ± 0.012	0.166 ± 0.014	0.585 ± 0.030	0.517 ± 0.008
MINs	0.746 ± 0.011	0.908 ± 0.004	0.323 ± 0.006	0.545 ± 0.031	0.519 ± 0.010
RoMA	0.298 ± 0.033	0.738 ± 0.018	0.391 ± 0.013	0.661 ± 0.010	0.525 ± 0.003
ICT	0.708 ± 0.019	0.898 ± 0.004	<u>0.431 ± 0.004</u>	0.667 ± 0.035	0.525 ± 0.016
Tri-mentoring	0.722 ± 0.015	0.902 ± 0.003	0.437 ± 0.005	0.683 ± 0.047	<u>0.531 ± 0.007</u>
LTR	0.679 ± 0.016	0.902 ± 0.002	0.421 ± 0.004	0.734 ± 0.025	0.508 ± 0.010
RaM-ListNet	0.717 ± 0.018	0.907 ± 0.003	0.400 ± 0.006	0.765 ± 0.028	0.518 ± 0.007
DDOM	0.749 ± 0.029	0.865 ± 0.009	0.419 ± 0.016	0.526 ± 0.017	0.506 ± 0.004
GTG	0.725 ± 0.077	0.913 ± 0.007	0.348 ± 0.017	0.611 ± 0.038	0.506 ± 0.006
GABO	0.016 ± 0.009	0.705 ± 0.002	0.344 ± 0.001	0.676 ± 0.021	0.512 ± 0.008
ROOT	0.847 ± 0.005	0.938 ± 0.002	0.397 ± 0.017	0.839 ± 0.015	0.526 ± 0.007
RG-EditFlow	<u>0.768 ± 0.014</u>	<u>0.915 ± 0.002</u>	0.410 ± 0.016	<u>0.836 ± 0.016</u>	0.516 ± 0.010

Table 3: Comparison with representative prior baselines on Design-Bench in terms of the normalized 50th-percentile score. The best and second-best results are shown in bold and underlined, respectively.

Method	Ant	D’Kitty	Superconductor	TFBind8	TFBind10
D_{best}	0.565	0.884	0.400	0.439	0.467
BO-qEI	0.569 ± 0.000	0.883 ± 0.000	0.300 ± 0.001	0.439 ± 0.000	0.469 ± 0.005
CMA-ES	-0.043 ± 0.007	0.674 ± 0.016	<u>0.374 ± 0.004</u>	0.536 ± 0.012	<u>0.490 ± 0.015</u>
REINFORCE	0.140 ± 0.026	0.510 ± 0.203	0.362 ± 0.006	0.450 ± 0.024	<u>0.470 ± 0.010</u>
COMs	0.471 ± 0.034	0.862 ± 0.003	0.306 ± 0.118	0.598 ± 0.031	0.475 ± 0.010
CbAS	0.369 ± 0.008	0.748 ± 0.016	0.101 ± 0.004	0.441 ± 0.021	0.465 ± 0.006
MINs	0.618 ± 0.016	0.889 ± 0.003	0.284 ± 0.011	0.421 ± 0.017	0.467 ± 0.010
RoMA	0.224 ± 0.020	0.545 ± 0.170	0.348 ± 0.019	0.519 ± 0.073	0.518 ± 0.003
ICT	0.554 ± 0.018	0.872 ± 0.007	0.371 ± 0.009	0.557 ± 0.031	0.457 ± 0.033
Tri-mentoring	0.572 ± 0.016	0.884 ± 0.001	0.404 ± 0.005	0.562 ± 0.051	0.475 ± 0.009
LTR	0.568 ± 0.016	0.885 ± 0.002	0.362 ± 0.004	0.566 ± 0.026	0.466 ± 0.011
RaM-ListNet	0.585 ± 0.012	0.888 ± 0.002	0.353 ± 0.006	0.556 ± 0.030	0.467 ± 0.008
DDOM	0.568 ± 0.066	0.814 ± 0.016	0.371 ± 0.017	0.404 ± 0.012	0.456 ± 0.002
GTG	<u>0.645 ± 0.098</u>	<u>0.901 ± 0.005</u>	0.284 ± 0.015	0.460 ± 0.032	0.452 ± 0.010
GABO	-0.039 ± 0.004	<u>0.674 ± 0.005</u>	0.299 ± 0.013	0.496 ± 0.011	0.457 ± 0.008
ROOT	0.712 ± 0.014	0.919 ± 0.003	0.348 ± 0.018	0.675 ± 0.026	0.473 ± 0.004
RG-EditFlow	0.630 ± 0.015	0.894 ± 0.001	0.372 ± 0.015	<u>0.625 ± 0.029</u>	0.468 ± 0.002

These percentile results provide a broader view of generated candidate quality. At the 80th percentile, RG-EditFlow is second-best on Ant, D’Kitty, and TFBind8, indicating that the proposed proposal-and-filter mechanism improves the upper portion of the candidate set on both morphology and sequence tasks. At the median, RG-EditFlow ranks second on TFBind8, while ROOT and GTG are stronger on several morphology-distribution metrics. Superconductor remains the clearest weak task, consistent with the main-paper result.

Appendix C: Training-Source Sensitivity

Table 4 keeps the inference protocol fixed and changes only the source of pseudo-edit supervision. *Recorded-path supervision* directly imitates recorded surrogate paths, whereas *kNN/mixup source* replaces GP-induced edit links with kNN/mixup pseudo-pairs. This isolates how the construction of local edit hypotheses affects the flow before the common trust-region reranker is applied.

Table 4: Training-source sensitivity across TFBind8, TFBind10, and D’Kitty. Entries are normalized mean / normalized 100th percentile over 8 evaluation seeds. Bold and underline mark the best and second-best 100th-percentile results per task.

Task	Full	Recorded-path supervision	kNN/mixup source
TFBind8	0.644 / 0.983	0.457 / <u>0.866</u>	0.439 / 0.802
TFBind10	0.471 / <u>0.649</u>	0.468 / 0.635	0.468 / 0.657
D’Kitty	0.891 / <u>0.962</u>	0.877 / 0.959	0.903 / 0.986

The results indicate that the usefulness of a pseudo-edit source depends on the task representation. On TFBind8, replacing the default GP-induced endpoint-pair supervision causes a large drop in both mean and 100th-percentile quality, suggesting that the default source captures score-directed sequence edits that the recorded-path and kNN/mixup alternatives do not recover. On TFBind10,

all three variants remain close, so the final quality is dominated more by inference-time filtering than by the exact training source. On D’Kitty, kNN/mixup improves the 100th percentile and slightly improves the mean, indicating that local interpolation between nearby morphology designs can provide competitive edit supervision for this continuous control task.

Appendix D: Endpoint Support for Training-Source Variants

Table 5 reports nearest-neighbor distance and seed displacement for the same training-source variants as Table 4. It asks whether changing the pseudo-edit source also changes where selected endpoints lie relative to the offline data support under the same inference protocol.

Table 5: Endpoint-support diagnostic for training-source variants. Lower nearest-neighbor distance and displacement indicate selected proposals closer to the offline support; all rows use the same inference protocol as the corresponding source-sensitivity result.

Variant	Mean	100th	NN dist.	Disp.
TFBind8 Full	0.644	0.983	6.69	7.85
TFBind8 Recorded path	0.457	0.866	2.61	1.86
TFBind8 kNN/mixup	0.439	0.802	2.04	1.75
TFBind10 Full	0.471	0.649	3.12	2.00
TFBind10 Recorded path	0.468	0.635	3.21	2.00
TFBind10 kNN/mixup	0.468	0.657	3.02	2.00
D’Kitty Full	0.891	0.962	1.85	1.00
D’Kitty Recorded path	0.877	0.959	1.90	1.00
D’Kitty kNN/mixup	0.903	0.986	1.65	1.00

For TFBind8, the recorded-path and kNN/mixup sources produce endpoints with much smaller nearest-neighbor distance and displacement than the full model, but their oracle quality is lower in Table 4. This shows that proximity to the offline support alone is insufficient; the edit source must also preserve score-improving direction. For D’Kitty, kNN/mixup is both closer to the offline support and higher-scoring, which is consistent with morphology edits benefiting from local interpolation among nearby designs. TFBind10 remains relatively stable across sources, matching the small quality differences in Table 4.

Appendix E: Task, Metric, and Hyperparameter Notes

Table 6 lists the task-level inference hyperparameters used for the five main-result runs. The notation follows the trust-region reranking equations in the main paper: M is the number of flow proposals per elite seed, r is the normalized seed-displacement clipping radius in the clipping equation, k is the number of offline nearest neighbors used by $\hat{s}_{\text{knn}}$ and U_{knn} , and (λ_u, λ_d) are the endpoint-level uncertainty and nearest-neighbor distance penalties in $R(\tilde{\mathbf{x}}; \mathbf{x}_0)$. We write $r = 0$ for TFBind8 because the implementation treats $r \leq 0$ as disabling clipping; this is the configuration used for the reported TFBind8 main result.

Table 6: Task-level inference hyperparameters for the main-result runs. All tasks use per-seed selection, label-variance uncertainty, and $Q = 128$ submitted candidates.

Task	M	σ_{prop}	r	k	(λ_u, λ_d)
Ant	16	0.005	1.0	5	(0.25, 0.50)
D’Kitty	16	0.005	1.0	5	(0.25, 0.50)
Superconductor	16	0.005	1.0	5	(0.25, 0.50)
TFBind8	16	0.010	0.0	5	(0.25, 0.10)
TFBind10	16	0.005	2.0	5	(0.25, 0.50)

The proposal noise σ_{prop} is applied only to the replicated initial states before ODE integration. All reranking quantities in Table 6 are computed from the offline dataset and the generated proposals before final oracle evaluation.